Reaction kinetics and pattern formation in a diffusive epidemic model with an infection-dependent recovery rate

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Sections:

1) Model and Mechanisms

2) ODE Dynamics

3) PDE Patterns

Location: Lombardy region, Italy
Disease: SARS-CoV-2 (COVID-19)

Events in chronological order:

Feb 20: First COVID-19 case reported in Italy.

Feb 23: Ten municipalities declared a “red zone.”

Mar 4: All schools are closed nationwide.

Mar 8: Fourteen northern provinces are placed under lockdown.

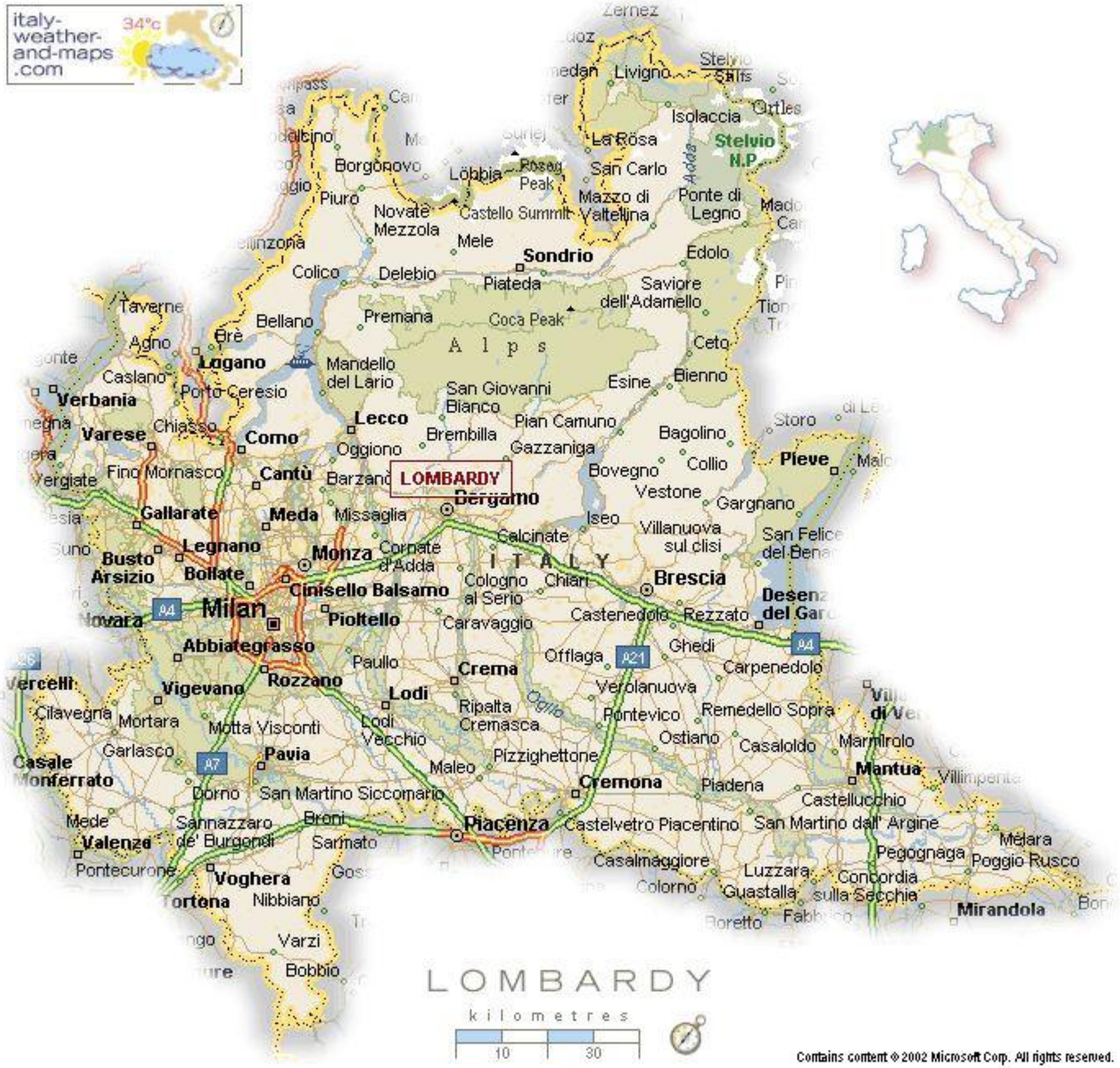
Mar 10: The lockdown is extended to all of Italy.

Mar 15: ICU capacity in Lombardy reaches full occupancy.

By late March 2020, Lombardy was Italy's COVID-19 epicenter.

Regional and national authorities moved quickly to expand ICU capacity by centralizing triage, transferring patients to less crowded hospitals, and opening the Fiera Milano field hospital.

Because Lombardy is tightly connected to nearby provinces through daily commuting and freight corridors, early transmission there helped seed outbreaks in adjacent areas.



Relief came from targeted actions that directly addressed the bottlenecks:

1) Increase ICU capacity and boost on-site care capabilities in the worst-hit zones.

2) Trim movement at hotspot nodes through transfers, staggered shifts, and reduced crowding on key routes.

Question: How can we include both the number of hospital beds and spatial spread in a mathematical model?

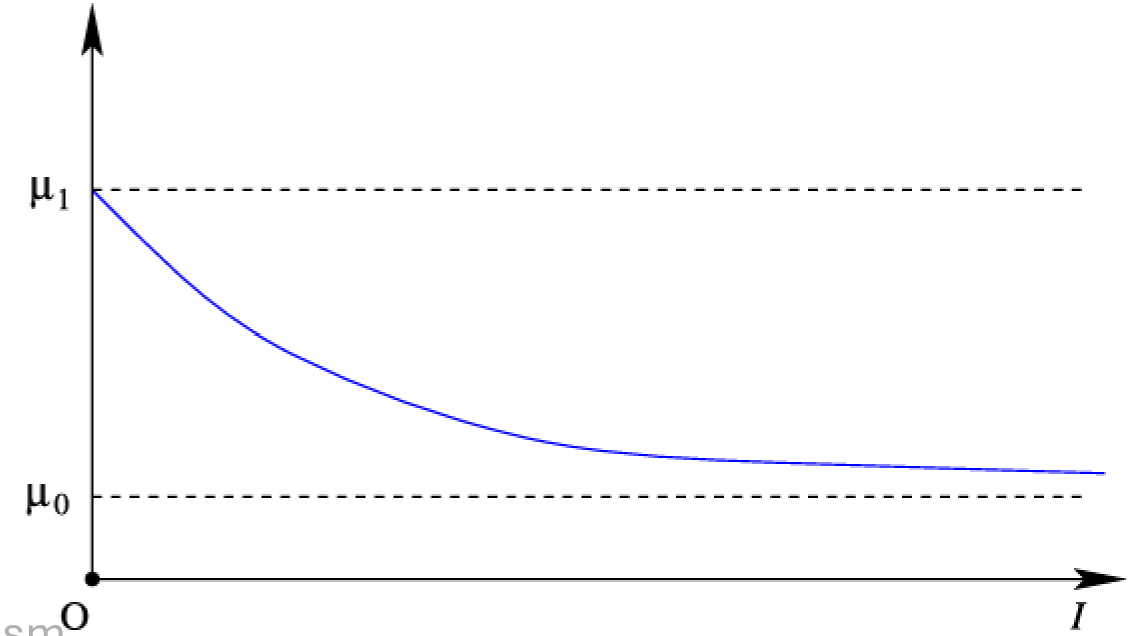
We start with a basic SIR model:

$$
\begin{aligned}
\frac{dS}{dt} &= A - d\,S - \beta SI, \\
\frac{dI}{dt} &= \beta SI - (d+\gamma)I, \\
\frac{dR}{dt} &= \gamma I - d\,R, \\
S(0) = S_0, &\quad I(0) = I_0, \quad R(0) = R_0.
\end{aligned}
\tag{1}
$$

1) Hospital Bed Availability: we introduce the function:

$$h(b, I) = \mu_0 + (\mu_1 - \mu_0)\frac{b}{I + b} \qquad (2)$$

Where h is the impact of hospital resource on the recovery rate, μ_0 and μ_1 are the minimum and maximum recovery rate per capita, b is the number of hospital beds and I is the number of infected individuals.



2) Spatial component: Fick's laws

Law 1 (flux): Flux moves from high to low concentration and is proportional to the gradient

$$\mathbf{J}\,(\mathrm{x},\mathrm{t}) = -D\,\nabla C(x, t).$$

Law 2 (conservation + Law 1):

$$\frac{\partial C}{\partial t} = D\,\Delta C.$$

If there are local sources or sinks $f(C, x, t),$ conservation gives the **reaction–diffusion** form:

$$\frac{\partial C}{\partial t} = D\,\Delta C + f(C, x, t).$$

Under the assumption of random non-biased moving of individuals, we arrive at the final model:

$$\begin{cases} S_t - r_1 \Delta S = A - dS - \beta SI, & x \in \Omega, t > 0; \\ I_t - r_2 \Delta I = \beta SI - dI - h(I)I, & x \in \Omega, t > 0; \\ R_t - r_3 \Delta R = h(I)I - dR, & x \in \Omega, t > 0; \\ S(x,0) = S_0(x), I(x,0) = I_0(x), R(x,0) = R_0(x), & x \in \Omega; \\ \frac{\partial S}{\partial \nu} = 0, \frac{\partial I}{\partial \nu} = 0, \frac{\partial R}{\partial \nu} = 0, & x \in \partial\Omega, t > 0, \end{cases} \tag{3}$$

The recovered class is assumed to have immunity over time, and that is why it does not return to the susceptible class, and the system can be decoupled as follows:

$$\begin{cases} S_t - r_1 \Delta S = A - dS - \beta SI, & x \in \Omega, t > 0; \\ I_t - r_2 \Delta I = \beta SI - dI - h(I)I, & x \in \Omega, t > 0; \\ S(x,0) = S_0(x) \geq 0, I(x,0) = I_0(x) \geq 0, & x \in \Omega; \\ \frac{\partial S}{\partial \nu} = 0, \frac{\partial I}{\partial \nu} = 0, & x \in \partial\Omega, t > 0. \end{cases} \tag{4}$$

Basic Analysis: ODE model

$$\begin{cases} \dfrac{dS}{dt} = A - dS - \beta SI, \\ \dfrac{dI}{dt} = \beta SI - dI - h(I)I. \end{cases} \tag{5}$$

Disease Free equilibrium

$$E_0 = \left(\frac{A}{d}, 0\right)$$

Endemic equilibrium

$$E_1 = (S_1, I_1)$$

Endemic equilibrium

$$E_2 = (S_2, I_2)$$

Disease Free equilibrium

$$E_0 = \left(\frac{A}{d}, 0\right)$$

By the method of next-generation matrix, the basic reproduction number is

$$\mathbb{R}_0 = \frac{\beta S}{d + h'(I)I + h(I)}\Big|_{(S,I)=(\frac{A}{d},0)} = \frac{\beta A}{d(d+\mu_1)}$$

Theorem 2.1. *Consider the local and global stability of E_0.*

(1) E_0 is locally asymptotically stable if $\mathbb{R}_0 < 1$ and unstable if $\mathbb{R}_0 > 1$.

(2) E_0 is globally asymptotically stable if $\beta A \leq d(d+\mu_0)$.

Proof:

For (1): Very direct

For (2): Consider the Lyapunov function I. The total derivative of I is

$$dI/dt = (\beta S - d - h(I))I \leq (\beta A/d - d - \mu_0)I \leq 0.$$

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For the endemic equilibria E_1 and E_2,

We have $S_i = \frac{h(I_i)+d}{\beta}$, and I_i is the solution of a quadratic equation

$$f(I) \coloneqq A_2 I^2 + A_1 I + A_0.$$

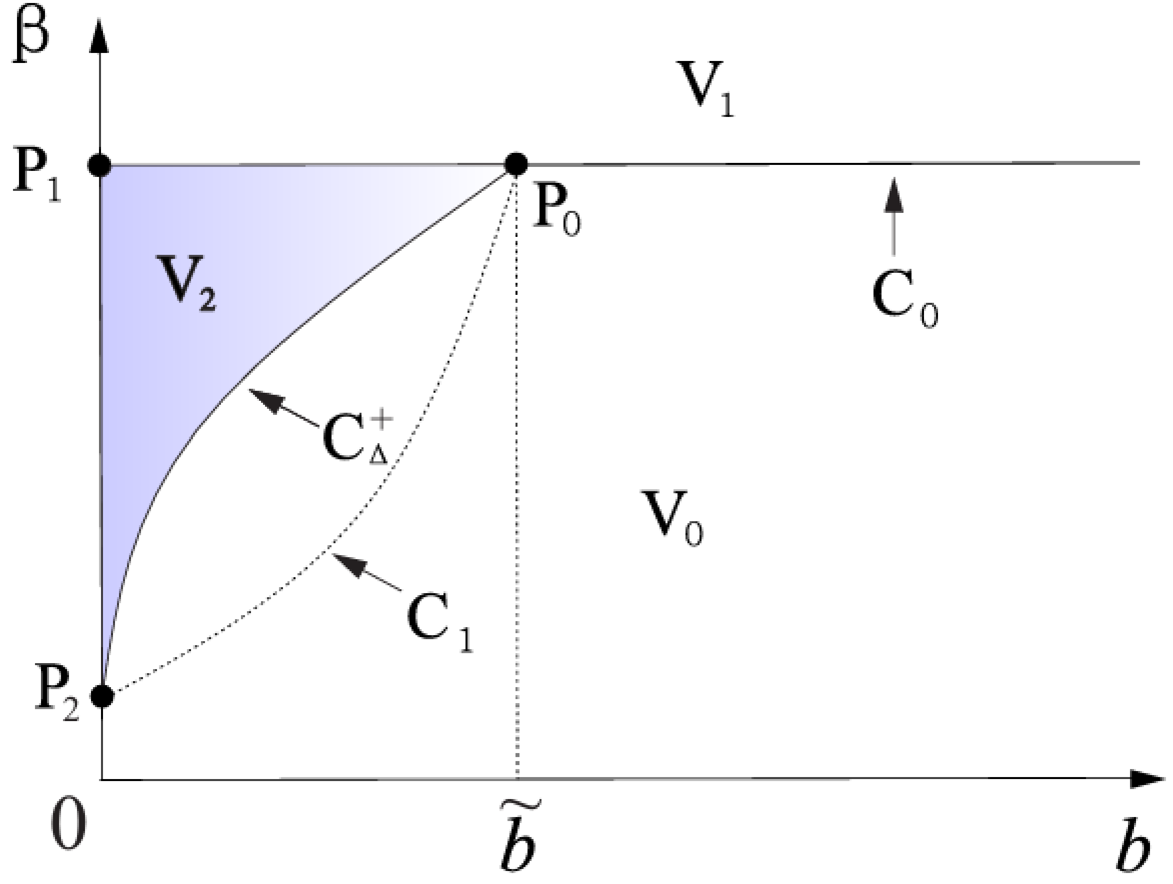
Define Δ to be the discriminant of $f(I) = 0$.

(a) *If* $\mathbb{R}_0 > 1$, *system* (5) *has a unique endemic equilibrium point* E_2.

(b) *If* $\mathbb{R}_0 = 1$ *and* $A_1 < 0$, *system* (5) *has a unique endemic equilibrium point* E_2.

(c) *If* $\mathbb{R}_0 < 1$, $\Delta > 0$ *and* $A_1 < 0$, *system* (5) *has two endemic equilibria* E_1 *and* E_2. *These two equilibria coalesce into* $E^*(S^*, I^*)$ *if* $\Delta = 0$.

(d) *System* (4) *does not have any endemic equilibrium point for other cases.*



- E_2 exists in V_1.
- E_1 and E_2 exist in V_2.
- No endemic equilibrium in V_0.
- E_1 disappears when crossing $\overline{P_0P_1}$ from V_2 to V_1.
- E_1 and E_2 coalesce into E^* on the arc $\overline{P_0P_2}$

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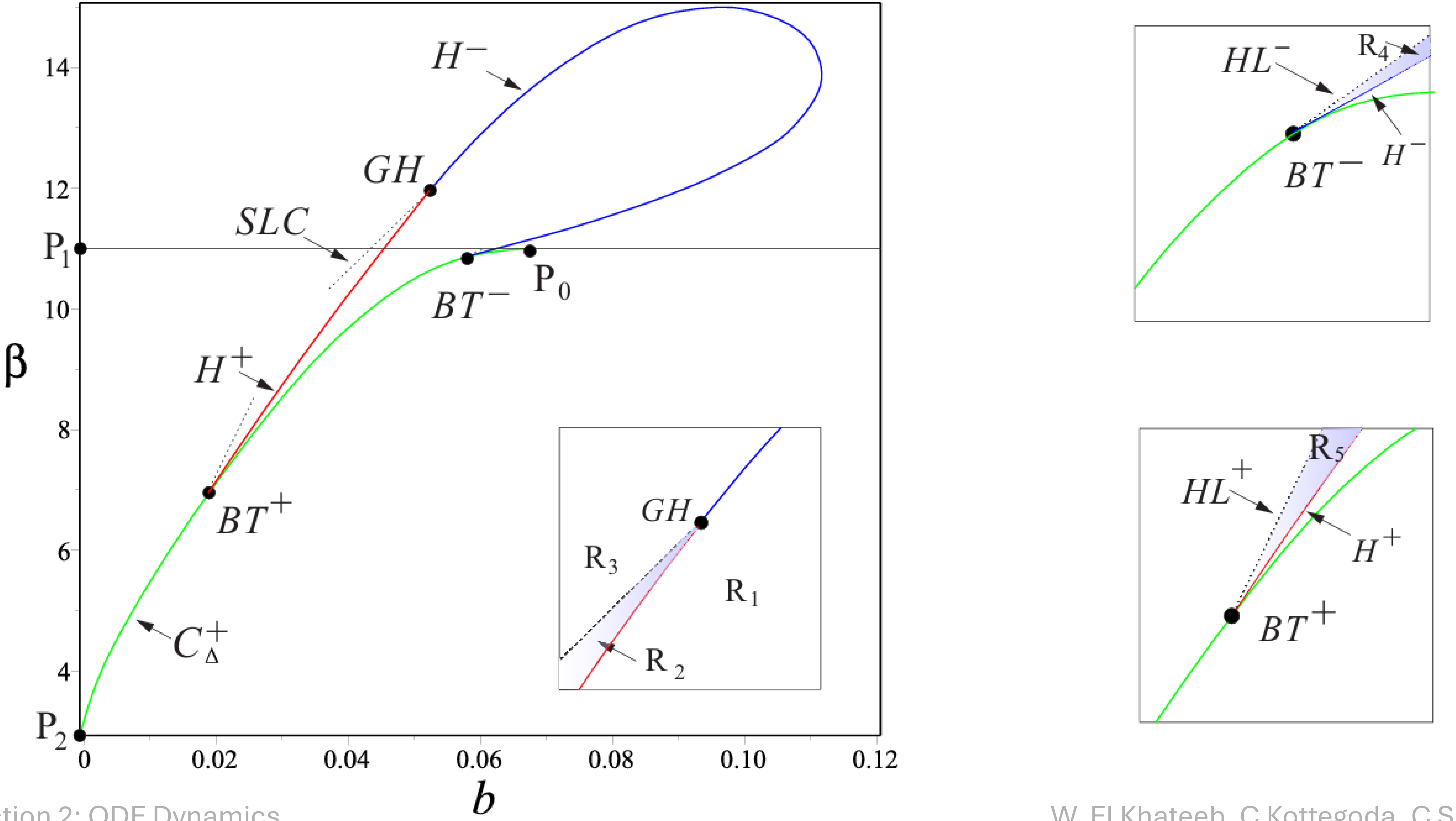
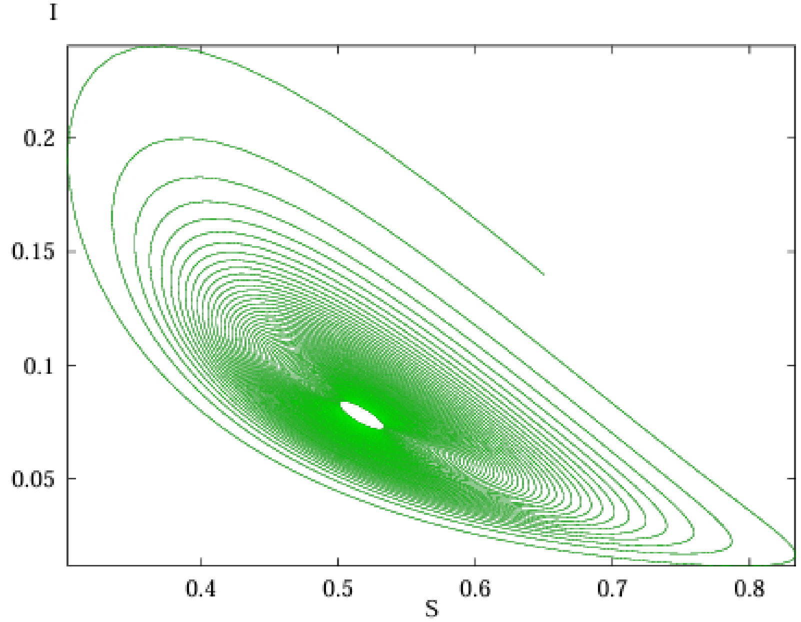
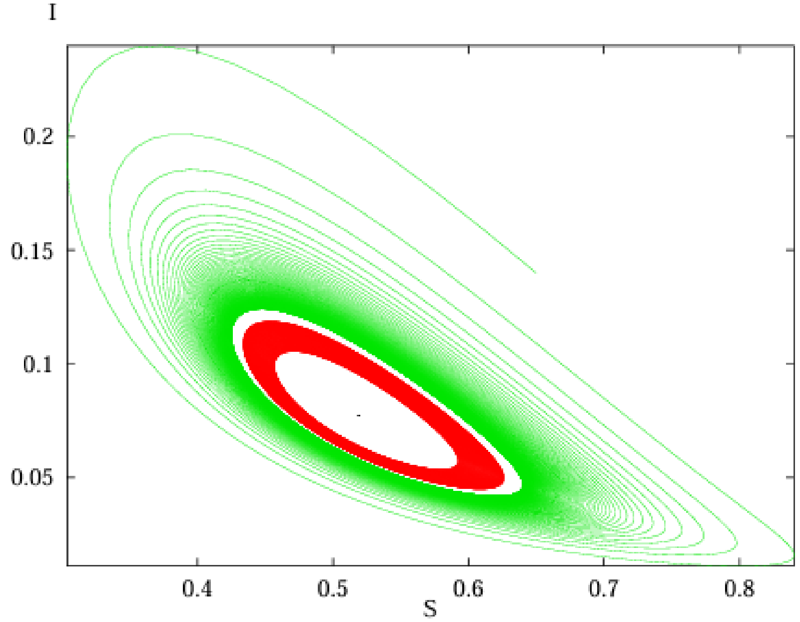


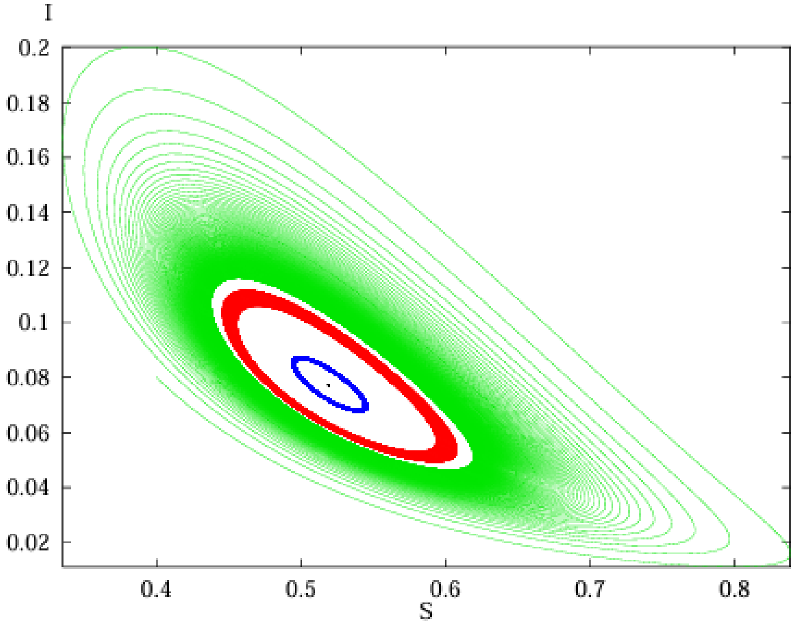
H⁻
GH
SLC
P1
P0
BT⁻
H⁺
BT⁺
C△⁺
β
b
P2
14
12
10
8
6
4
0
0.02
0.04
0.06
0.08
0.10
0.12
R3
R1
R2
HL⁻
R4
BT⁻
HL⁺
R5



(a) E_2 is stable and no periodic solution in R_3



(b) One periodic solution in R_1



(c) Two periodic solutions in R_2

We Introduce the linear operator $\mathcal{L}$ on the Sobolov Space X_C as

$$\mathcal{L}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix} \begin{bmatrix} u_{xx} \\ v_{xx} \end{bmatrix} + J_0(\bar{S}, \bar{I}) \begin{bmatrix} u \\ v \end{bmatrix}, \qquad (6)$$

where $(\bar{S}, \bar{I})$ is an equilibrium point of the underlying ODE system and $J_0(\bar{S}, \bar{I})$ is the Jacobian of system at $(\bar{S}, \bar{I})$. Then the linearization of the reaction-diffusion system at $(\bar{S}. \bar{I})$ is

$$\begin{bmatrix} u_t \\ v_t \end{bmatrix} = \mathcal{L}\begin{bmatrix} u \\ v \end{bmatrix}. \qquad (7)$$

Let Z^+ be the set of all positive integers. Given the homogeneous Neumann boundary condition, the Fourier series solution of the linear system takes the form

$$\begin{bmatrix} u(x,t) \\ v(x,t) \end{bmatrix} = \sum_{k=0}^{\infty} e^{\lambda t} \cos \frac{k}{\ell} x \begin{bmatrix} \phi_k \\ \psi_k \end{bmatrix}, \qquad (8)$$

where $k \in Z^+ \cup \{0\}$, while $\frac{k}{\ell}$ is known as the wavenumber which indicates how rapidly the oscillations occur in space, with higher values corresponding to shorter wavelengths (i.e., finer spatial structures)

Finally, λ is the eigenvalue of

$$J_k(\bar{S},\bar{I}) = J_0(\bar{S},\bar{I}) - \frac{k^2}{\ell^2}\begin{bmatrix} r_1 & 0 \\ 0 & r_2 \end{bmatrix}, \quad k \in \mathbb{Z}^+ \cup \{0\}. \tag{9}$$

Theorem 3.1. *The constant steady state* $E_0(\frac{A}{d},0)$ *of system* (1.2) *is stable if* $\mathbb{R}_0 < 1$ *and unstable if* $\mathbb{R}_0 > 1$.

$$J_k(\frac{A}{d},0) = \begin{bmatrix} -(d+\frac{r_1k^2}{\ell^2}) & -(d+\mu_1)\,\mathbb{R}_0 \\ 0 & (d+\mu_1)(\mathbb{R}_0-1) - \frac{r_2k^2}{\ell^2} \end{bmatrix}. \tag{10}$$

There are two eigenvalues $\lambda_1 = -(d+\frac{r_1k^2}{\ell^2}) < 0$ and $\lambda_2 = (d+\mu_1)(\mathbb{R}_0-1) - \frac{r_2k^2}{\ell^2}$. If $\mathbb{R}_0 < 1$, $\lambda_2 < 0$ fro all $k \in \mathbb{Z}^+ \cup \{0\}$, then E_0 is stable. If $\mathbb{R}_0 > 1$, then $\lambda_2 > 0$ for $k = 0$, and E_0 is unstable. $\square$

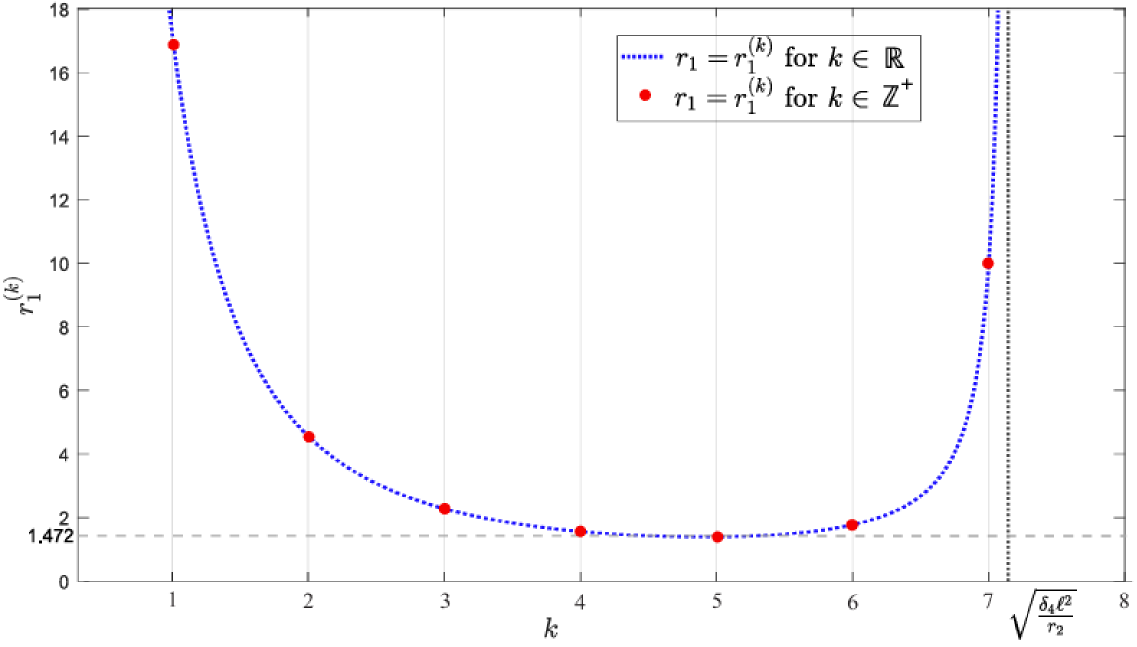
Definition 3.3. *[25] Let $k \in \mathbb{Z}^+$ be a positive integer and fixed. Let $\xi \in \mathbb{R}$ be a bifurcation parameter. A k-mode Turing bifurcation occurs if the Jacobian J_k has a unique real eigenvalue $\lambda = \lambda(k)$ crossing zero transversally provided that the spectral gap condition holds, that is, all remaining eigenvalues of $J_n(n = 0, 1, \cdots)$ are to the left of the imaginary axis as ξ passing through $\xi = \xi^* \in \mathbb{R}$.*

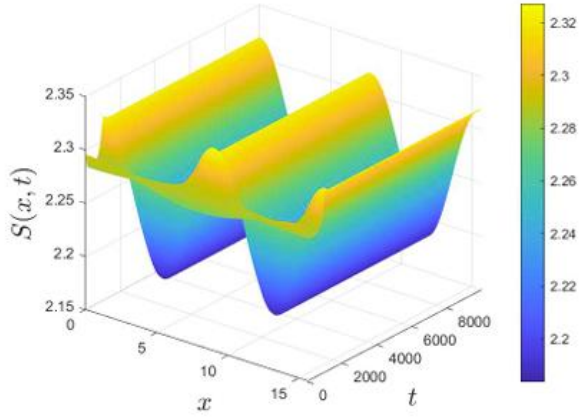
To study the instability caused by the diffusion on the steady state E_2, we write the determinant of $J_k(E_2)$ as:

$$\mathcal{D}_k = \left[\left(\frac{k}{\ell}\right)^2 r_2 - \delta_4\right]\left(\frac{k}{\ell}\right)^2 r_1 - \left(\frac{k}{\ell}\right)^2 \delta_1 r_2 + \delta_1\delta_4 - \delta_2\delta_3.$$

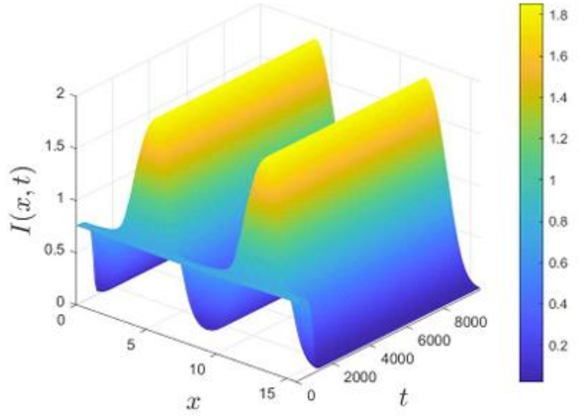
Which vanishes for:

$$r_1 = r_1^{(k)} := \frac{(\delta_1\delta_4 - \delta_2\delta_3 - \delta_1 r_2 \frac{k^2}{\ell^2})}{\frac{k^2}{\ell^2}(\delta_4 - r_2\frac{k^2}{\ell^2})}$$

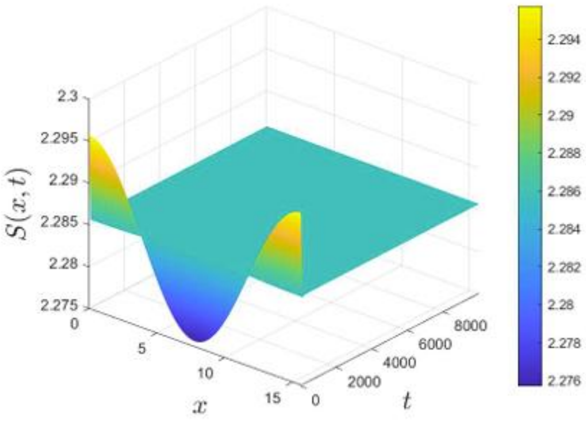




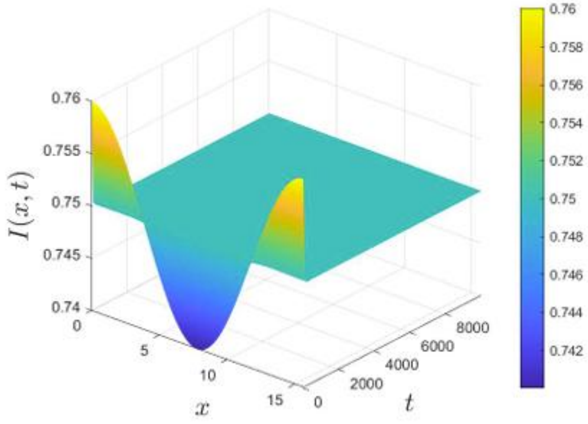
(a) Surface $S(x,t)$



(b) Surface $I(x,t)$

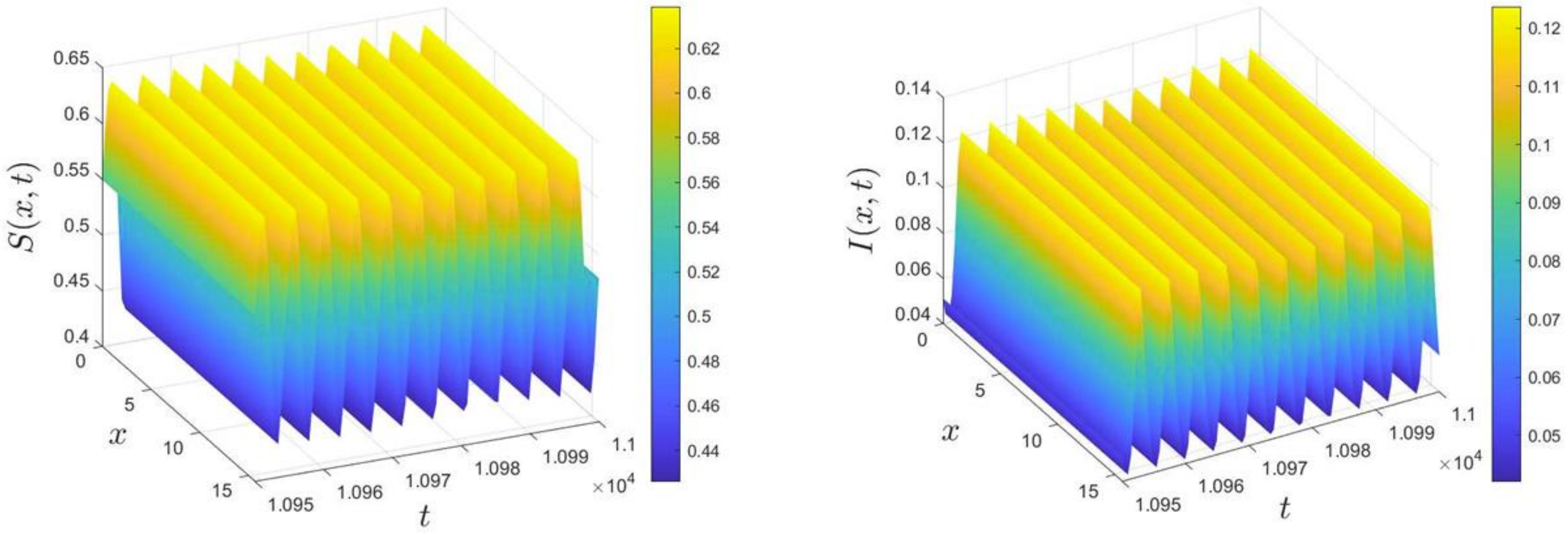


(a) Surface $S(x,t)$



(b) Surface $I(x,t)$

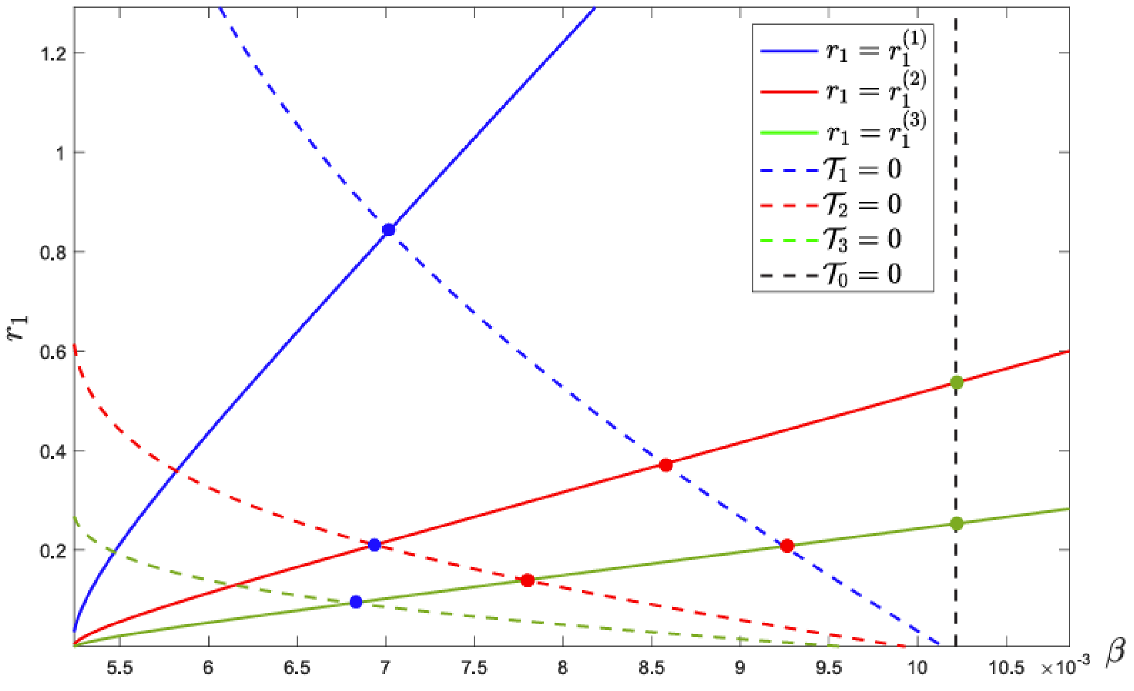
Definition *Let $k \in \mathbb{Z}^+ \cup \{0\}$ be an integer and fixed. Let $\xi \in \mathbb{R}$ be a bifurcation parameter. A k-mode Hopf bifurcation occurs if the Jacobian J_k has a pair of purely imaginary eigenvalues crossing the imaginary axis transversally and all eigenvalues of $J_n (n \neq k)$ are not on the imaginary axis as ξ passes through $\xi = \xi^* \in \mathbb{R}$.*

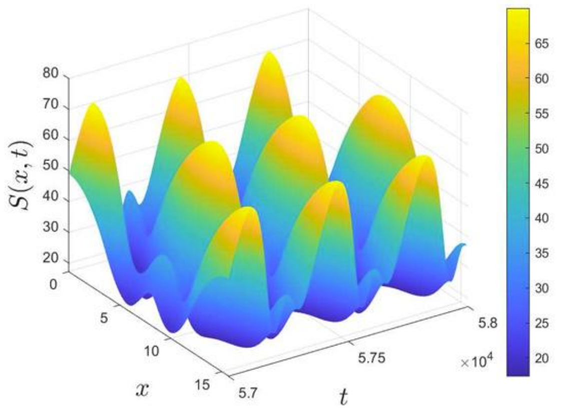


(a) Surface $S(x, t)$ (b) Surface $I(x, t)$

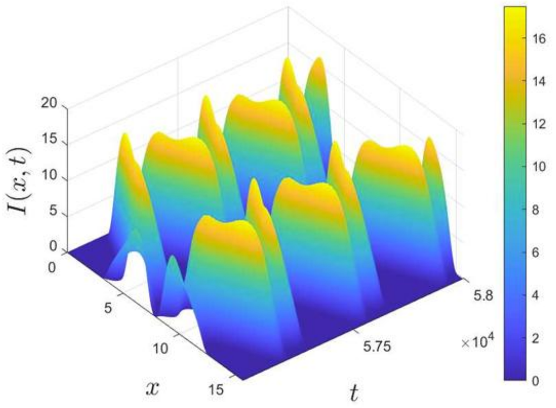
Figure 7: A spatially homogeneous periodic solution generated by 0-mode Hopf bifurcation. It is the spatiotemporal version of the limit cycle in Fig. 3 (b).

(k_1, k_2) –Turing Hopf Bifurcation

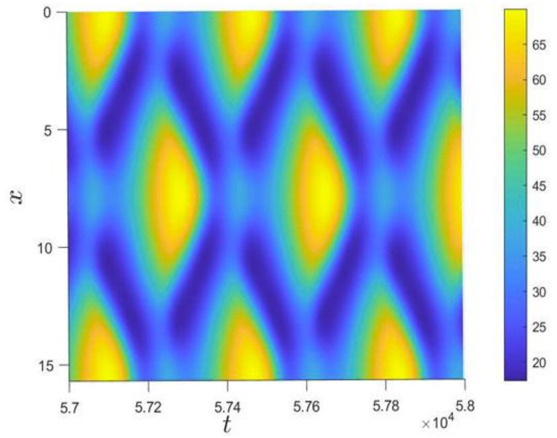




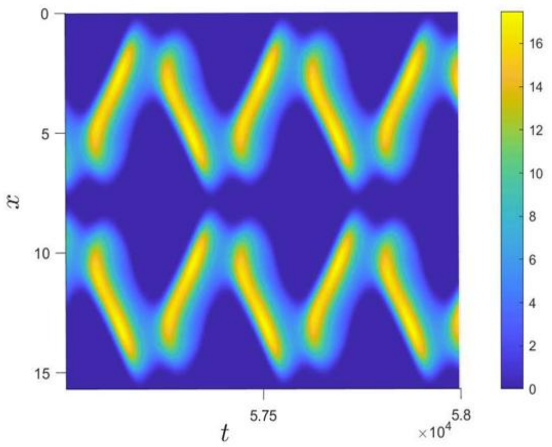
(a) Surface $S(x,t)$



(b) Surface $I(I,t)$

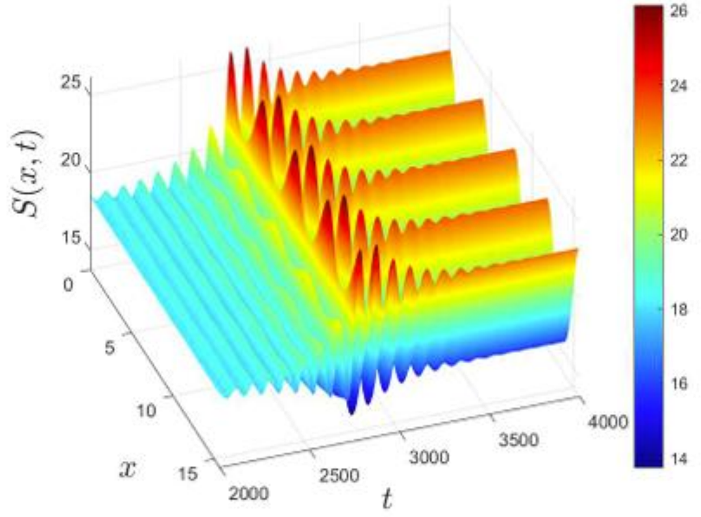


(a) Projection of $S(x,t)$

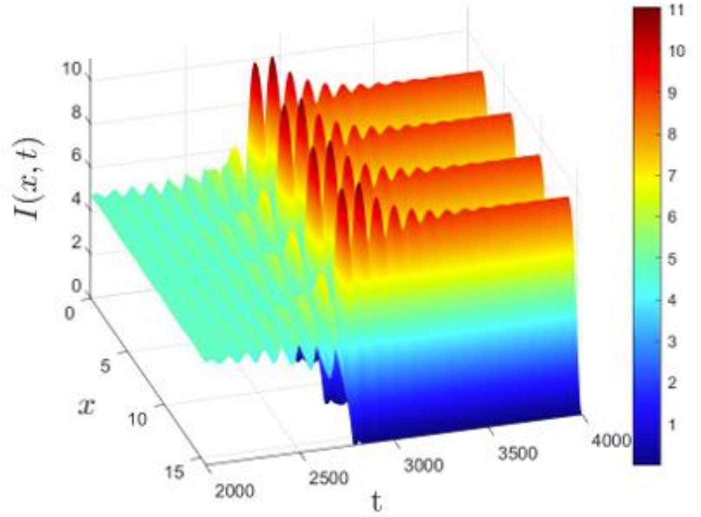


(b) Projection of $I(x,t)$

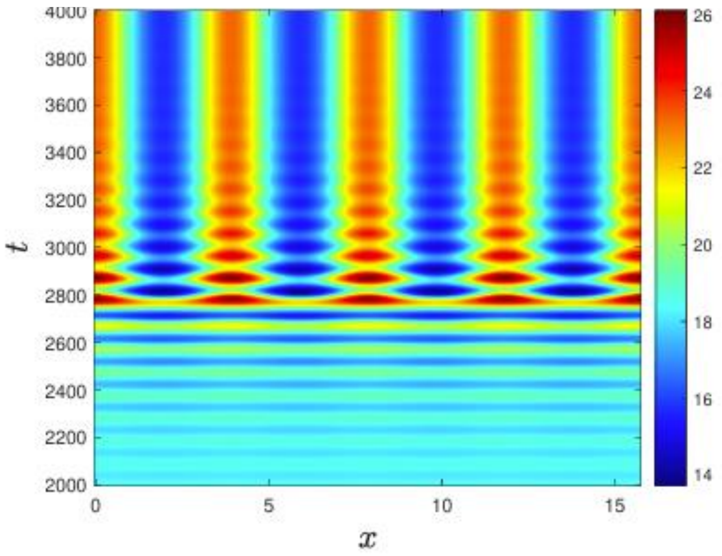
Transient dynamics from temporal oscillatory regime to a spatially periodic pattern



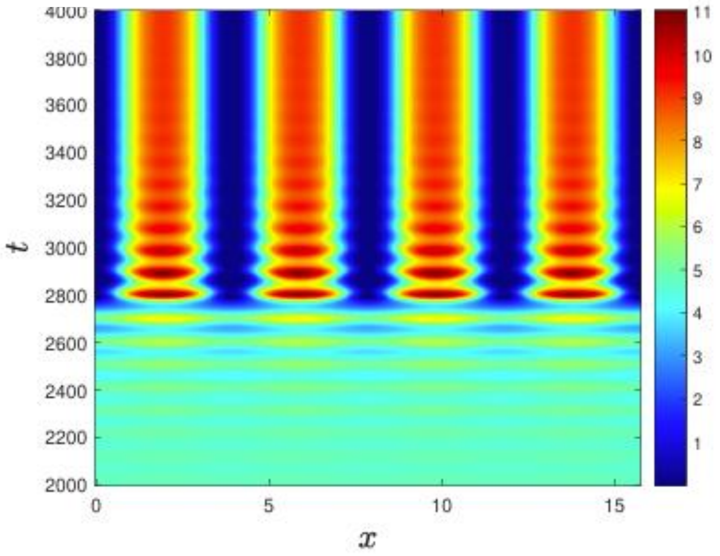
(a) Surface of $S(x,t)$



(b) Surface of $I(x,t)$



(a) Projection of $S(x,t)$



(b) Projection of $I(x,t)$

Transient dynamics observed:

•Early stage ($t < 2800$): temporally periodic, spatially homogeneous oscillations.

•Post-transition ($t > 2800$): shift to spatiotemporal patterns with localized hotspots and wave structures.

Public Health implication:

- **Uniform interventions are insufficient:**

Strategies applied evenly across space (e.g., mass vaccination, broad policies) may not control outbreaks effectively.

- **Spatial targeting is essential:**

Interventions must adapt to **regional heterogeneity** and localized hotspots.

- **Cyclic and regional disease waves:**

Control policies should account for **asynchronous recurrence** of infection across different areas.

- **Adaptive strategy required:**

Combining **temporal timing** (when to intervene) with **spatial targeting** (where to intervene) is key to containment.

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